

# Electronic Voting System

## Input, Output, and Report Design Document

### 1. Introduction

The Electronic Voting System is designed to automate election activities such as voter registration, candidate management, vote recording, polling station management, and election result generation. The system improves election accuracy, security, transparency, and efficiency.

This document presents:

- Input Design
- Output Design
- Report Design

### 2. Input Design

Input design focuses on how data is entered into the system. Proper input design helps reduce errors, improve data accuracy, and simplify system usage.

#### 2.1 Voter Registration Form

##### Purpose

Used to register voters into the system.

##### Sample Form Design

| VOTER REGISTRATION |                  |
|--------------------|------------------|
| Full Name          | : [_____]        |
| Date of Birth      | : [__]/[__]/[__] |
| Gender             | : [_____]        |
| Address            | : [_____]        |
| National ID        | : [_____]        |
| Phone Number       | : [_____]        |
| Email              | : [_____]        |
| Region             | : [_____]        |
| Status             | : [_____]        |

##### Input Table

| Field Name    | Type        | Description        |
|---------------|-------------|--------------------|
| voter_id      | Integer     | Primary Key        |
| full_name     | Text        | Voter full name    |
| date_of_birth | Date        | Birth date         |
| gender        | Text        | Male or Female     |
| address       | Text        | Home address       |
| national_id   | Text        | Unique national ID |
| phone_number  | Text        | Contact number     |
| email         | Text        | Email address      |
| region_code   | Foreign Key | Region reference   |
| status        | Text        | Active/Inactive    |

## 2.2 Candidate Registration Form

### Purpose

Used to register election candidates.

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### Sample Form Design

#### CANDIDATE REGISTRATION

|                                         |                |
|-----------------------------------------|----------------|
| Candidate Name : [_____]                |                |
| Date of Birth : [___/___/___]           |                |
| Gender                                  | : [ Male v ]   |
| Address : [_____]                       |                |
| Political Party: [ Prosperity Party v ] |                |
| Status                                  | : [ Active v ] |
| Photo Upload : [ Choose File ]          |                |

### Input Table

| Field Name   | Type    | Description |
|--------------|---------|-------------|
| candidate_id | Integer | Primary Key |

|                       |             |                  |
|-----------------------|-------------|------------------|
| <b>candidate_name</b> | Text        | Candidate name   |
| <b>date_of_birth</b>  | Date        | Birth date       |
| <b>gender</b>         | Text        | Gender           |
| <b>address</b>        | Text        | Address          |
| <b>party_id</b>       | Foreign Key | Party reference  |
| <b>status</b>         | Text        | Candidate status |
| <b>photo_url</b>      | Text        | Candidate photo  |

## 2.3 Political Party Form

### Purpose

### Input Table

| Field Name                 | Type    | Description                |
|----------------------------|---------|----------------------------|
| <b>party_id</b>            | Integer | Primary Key                |
| <b>party_name</b>          | Text    | Party name                 |
| <b>party_abbreviation</b>  | Text    | Short name                 |
| <b>registration_number</b> | Text    | Unique registration number |
| <b>description</b>         | Text    | Party description          |
| <b>symbol_url</b>          | Text    | Party logo                 |

## 2.4 Polling Station Form

### Purpose

Used to register polling stations.

### Sample Form Design

#### POLLING STATION FORM

Station Name : [\_\_\_\_\_]

Station Code : [\_\_\_\_\_]

Address : [\_\_\_\_\_]

Latitude : [\_\_\_\_\_]

Longitude : [\_\_\_\_\_]

Region : [\_\_\_\_\_]

### Input Table

| Field Name                | Type        | Description      |
|---------------------------|-------------|------------------|
| <b>polling_station_id</b> | Integer     | Primary Key      |
| <b>station_name</b>       | Text        | Station name     |
| <b>station_code</b>       | Text        | Unique code      |
| <b>address</b>            | Text        | Station address  |
| <b>latitude</b>           | Decimal     | GPS latitude     |
| <b>longitude</b>          | Decimal     | GPS longitude    |
| <b>region_code</b>        | Foreign Key | Region reference |

## 2.5 Voting Form

### Purpose

Used by voters to cast votes.

### Input Table

| Field Name                | Type        | Description     |
|---------------------------|-------------|-----------------|
| <b>vote_id</b>            | Integer     | Primary Key     |
| <b>voter_id</b>           | Foreign Key | Voter reference |
| <b>party_id</b>           | Foreign Key | Selected party  |
| <b>polling_station_id</b> | Foreign Key | Polling station |

|                  |          |               |
|------------------|----------|---------------|
| <b>vote_date</b> | DateTime | Voting time   |
| <b>is_valid</b>  | Boolean  | Vote validity |

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### 3. Output Design

Output design explains how processed information is displayed to users.

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#### 3.1 Voter Information Output

##### Sample Output

##### Voter information

Voter ID : 1001

Full Name : abebe balcha

National ID : ET123456

Region : Addis Ababa

Status : Active

#### 3.2 Candidate Information Output

##### Sample Output

Candidate Name : Kebede mola

Political Party: prosperity Party

Status : Active

#### 3.3 Election Result Output

##### Sample Output

| Party Name | Total Votes | Percentage |
|------------|-------------|------------|
| Party A    | 15,000      | 45%        |
| Party B    | 12,000      | 36%        |
| Party C    | 6,000       | 19%        |

#### 3.4 Polling Station Output

## Sample Output

| Station Code | Station Name     | Region      |
|--------------|------------------|-------------|
| PS001        | Addis Station 01 | Addis Ababa |
| PS002        | Adama Station 02 | Oromia      |

## 3.5 Notification Messages

### Example Messages

✓ Registration Completed Successfully

✓ Vote Submitted Successfully

✗ Invalid Username or Password

✗ Duplicate Vote Detected

## 4. Report Design

Reports are generated to support election management and monitoring.

### 4.1 Registered Voters Report

#### Purpose

Displays all registered voters.

#### Sample Report Table

| Voter ID | Full Name     | Region      | Status |
|----------|---------------|-------------|--------|
| 1001     | Dawit Aregawi | Addis Ababa | Active |
| 1002     | Abel Tesfaye  | Oromia      | Active |

### 4.2 Election Result Report

#### Purpose

Displays election vote results.

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#### Sample Report Table

| Party   | Votes  | Percentage |
|---------|--------|------------|
| Party A | 15,000 | 45%        |
| Party B | 12,000 | 36%        |
| Party C | 6,000  | 19%        |

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### 4.3 Candidate Report

#### Sample Report Table

| Candidate Name | Party          | Status |
|----------------|----------------|--------|
| Abebe Kebede   | Unity Party    | Active |
| Hana Bekele    | National Party | Active |

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### 4.4 Polling Station Report

#### Sample Report Table

| Station Name | Region      | Address |
|--------------|-------------|---------|
| Station 01   | Addis Ababa | Bole    |
| Station 02   | Oromia      | Adama   |

### 4.5 Invalid Vote Report

#### Sample Report Table

| Vote ID | Voter ID | Reason            |
|---------|----------|-------------------|
| 5001    | 1002     | Duplicate Vote    |
| 5002    | 1045     | Invalid Candidate |

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### 4.6 Admin Activity Report

#### Sample Report Table

| Admin Name | Activity                | Date       |
|------------|-------------------------|------------|
| Admin01    | Registered Voter        | 2026-05-01 |
| Admin02    | Updated Polling Station | 2026-05-02 |

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## 5. Conclusion

The Electronic Voting System input, output, and report designs help ensure efficient election processing, proper data management, accurate reporting, and improved transparency. The structured forms and reports make the system user-friendly and suitable for real-world election management.

The input, output, and report designs of the Electronic Voting System are developed to ensure efficient election management, accurate data processing, and secure handling of voter information. Properly designed interfaces and reports improve usability, maintain data integrity, and support transparent election operations.